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PUBLIC REVIEW DRAFT:
DUNE Interface Document:
Slow Controls – Ground Impedance Monitor (GIZMo)

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Summary

This public review copy describes the interface between DUNE Slow Controls (SC) and the Ground Impedance Monitor (GIZMo). SC supports two hardware implementations: GIZMo Kria, built around an AMD/Xilinx Kria K26 SOM on the KR260 Robotics Starter Kit carrier, and legacy GIZMo, built around the ZedBoard. The Kria implementation defines the common supervisory OPC UA contract; the separate ZedBoard server conforms to it while retaining different internal hardware, firmware, operating-system, and control capabilities. The document separates the physical measurement layer, hardware-control layer, internal application protocols, public OPC UA interface, Ignition display and history, Detector Protection boundary, calibration authority, and commissioning gates.

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Distribution List

GIZMo system owner; DUNE Slow Controls; Detector Protection System; detector operations; installation and integration; controls cybersecurity; Ignition support.

History of Changes

Revision	Date	Author	History of changes
0.1	2026-08-12	M. Arroyave	Initial dual-platform draft and companion DCS intake spreadsheets.
0.2	2026-08-12	M. Arroyave	Reorganized as a layered interface-control document following the SC–PDS precedent. Distinguished native controller capability from presently exposed SC control and added the legacy controlled- expansion gates.
0.3	2026-08-12	M. Arroyave	Defined GIZMo Kria and legacy GIZMo as two supported hardware implementations, identified the K26 SOM/KR260 and ZedBoard platforms, and removed hardware-performance-precedence requirements.
0.4	2026-08-12	M. Arroyave	Audited all workbook numeric fields against the maintained Kria runtime, recovered legacy behavior, engineering records, and commissioning observations. Defined the then-proposed initial threshold, alarm timing, and removed inherited template example values.
0.5	2026-08-12	M. Arroyave	Set the initial threshold to 100 ohm for both implementations. Used the retained Kria SQLite history and read-only live samples from the connected ZedBoard to populate clearly labelled commissioning values and operating envelopes; confirmed that no OPC UA server currently runs on the ZedBoard.

0.6	2026-08-13	M. Arroyave	Implemented and deployed the initial monitoring-only off-board ZedBoard adapter. Added pinned target identity, hash-locked coherent RUN acquisition, explicit non-good unsupported values, and independent rejection of all OPC UA writes and methods.
0.7	2026-08-13	M. Arroyave	Delivery review synchronized document and workbook revision metadata, reconciled the selected signal attributes with the canonical model, completed identity and health summary rows, and incorporated the partial 27-cycle legacy-adapter pilot without closing its acceptance gate.
0.7-public	2026-08-13	M. Arroyave	Created a clearly labelled public review copy with site endpoints, executable fingerprints, access details, and controlled source records omitted.
0.8-public	2026-08-13	M. Arroyave	Added the public human-facing connection, power-up, and construction-monitoring guide; identified the planned legacy OPC UA server as the production integration dependency while retaining the off-board adapter pilot as commissioning evidence; and aligned history guidance with direct Ignition OPC UA subscription and tag history.
0.9-public	2026-08-13	M. Arroyave	Recorded a target-side development build of the native ZedBoard OPC UA server, recovery journal, and automatic time service. Exercised authenticated `Configuration.ThresholdOhm` control over the recovered 0–1023-ohm range in an idempotent 100-ohm end-to-end test; retained all other legacy settings and methods as read-only or explicitly unsupported.

0.10-public	2026-08-14	M. Arroyave	Made the generated Kria model 1.3.1 schema the authoritative GIZMo–SC contract and classified the ZedBoard server as a separate conforming implementation in development. Preserved canonical NodeIds, datatypes, units, and the 0–1,000,000-ohm threshold metadata; recorded 0–1023 ohm only as the ZedBoard accepted subset; and prohibited proxying, cross-device control, and competing ownership of legacy hardware. Also clarified that `HIGHZ` is a valid good-quality range state with no fabricated numeric resistance.

Contents

1 Introduction

The GIZMo monitors the impedance relationship between detector ground and building ground. It applies an AC stimulus, observes the returned signal through the installed current-transformer and analog path, calculates impedance-related quantities, and provides a local alarm indication. SC uses the public machine interface for authorized configuration and operations, operator display, alarms, and history.

The maintained GIZMo runtime (package version 0.4.6, OPC UA model 1.3.1) provides the native Kria endpoint and the generated machine-readable contract. The legacy GIZMo runs the preserved 2017 ARM controller; a separate, capability-bounded OPC UA server is under development for its ZedBoard. SC shall support both implementations through independent connections. Earlier monitoring-adapter and target-side server tests remain development evidence, not production acceptance. The recovery collector and conforming server shall validate the controller against a controlled SHA-256 fingerprint before publishing data. The fingerprint is omitted from this public copy and retained in the commissioning record:

```
<controlled-executable-sha256>
```

This ICD specifies interfaces and ownership; it does not replace electrical schematics, the device software design, a Detector Protection design, or a traceable metrology procedure. In particular, existence of legacy calibrator hardware and a `CAL` command does not establish a production self- calibration process.

1.1 Applicable and reference documents

Document	Use in this ICD
DUNE-doc 1805 v.12, *Ground Impedance Monitor* documentation set	Legacy GIZMo engineering and setup documentation, including *Ground Impedance Monitor Engineering Note 170621*, *Impedance Monitor System Setup Instructions 170621*, *ZMon System Diagram 170616*, ZMon Amplifier Board drawing 176962, and the ZedBoard Rev D schematic.
DUNE-doc 27482 v.2, *Ground Impedance Monitor* test, calibration, and chassis documentation set	Legacy PMod ZMon Calibrator drawing 176971, revised amplifier and chassis drawings, test procedures, SD-card procedure, and recorded characterization data.
DUNE-doc 35392 v.2, *ORC Documents for DUNE Impedance Monitor Rack in FNAL HEERC G271*	Kria GIZMo rack, power, equipment, safety, and ORC context.

GIZMo-Kria, T. DeLine, 26 June 2025; and *GIZMo-Kria System Development/User Manual*, T. DeLine, 22 July 2026	Kria hardware, firmware, software, connections, startup, operator access, and calibration guidance.
Maintained GIZMo runtime 0.4.6 documentation, including the generated `schema/gizmo-opcua-contract.json`, `docs/opcua.md` and `docs/architecture.md`	Implemented GIZMo Kria authoritative OPC UA contract, service boundaries, and control behavior.
DUNE-doc 37315 v.1, *Communication Architecture for DUNE FD Slow Controls*	Current SC bridge-application, OPC UA, Ignition, protocol-hook, and hardware-emulator direction.

1.2 Definitions and requirement language

Term	**Definition**
GIZMo Kria	GIZMo implementation using an AMD/Xilinx Kria K26 SOM on the KR260 Robotics Starter Kit carrier.
Legacy GIZMo	GIZMo implementation using the ZedBoard and the preserved 2017 controller software.
Contract authority	The Kria implementation and its generated `urn:fnal:gizmo` model schema. It defines the required NodeIds, datatypes, access metadata, engineering metadata, methods, and meanings used by SC.
Conforming producer	An independent OPC UA server that implements the authoritative contract without narrowing or reinterpreting it. A platform may reject a locally unsupported write or method with the appropriate status.
Native capability	A function present in the platform software or service manager, whether or not it is exposed to SC.
Exposed control	A validated, authorized operation available through the public OPC UA contract.
Reset	An ambiguous word that shall be qualified as *clear alarm latch*, *restart measurement engine*, *reboot controller*, or *restore calibrator high-impedance state*. These actions are not interchangeable.

Authoritative alarm	Alarm decision produced by the device branch that controls its physical alarm outputs; not a state recreated by SC.
HIGH Z	Resistance outside the validated measurement range. It is a state, not a fabricated numeric endpoint.
CharacterizationOnly	Hardware or historical sweep data exist, but no production calibration has completed validation and controlled activation.

"Shall" denotes a normative requirement. "Should" denotes a recommended implementation. Open values are marked TBD and shall not be interpreted as zero, disabled, or accepted by omission.

2 System Architecture

2.1 Layered architecture

```
Layer 0  Detector ground <-> building ground / CT / saturable inductor
                           |
Layer 1  GIZMo analog front end, ADC/DAC, FPGA, relays, local display
                           |
Layer 2  Platform controller and private application protocols
         Kria: ZMon :5055, ZMQ :5555, private control socket
         Zed:  GIZMO.elf, FPGA MMIO, VM801P SPI, SysV service
                           |
Layer 3  Public interface: OPC UA Binary over UA TCP
                           |
Layer 4  Ignition: tags, controls, alarms, trends, and history
```

Only Layer 3 is the normal SC command/data boundary. Interfaces in Layer 2 are private implementation details and shall not become independent SC control paths. Layer 0 and Layer 1 work is a controlled electrical or calibration activity.

2.2 Implementation topology

Hardware	Acquisition and control path	SC consumers
GIZMo Kria	K26 SOM/KR260 measurement engine $\rightarrow$ native canonical OPC UA endpoint	Ignition tags, controls, alarms, trends, and history

Legacy GIZMo	ZedBoard `GIZMO.elf` → target-local recovery journal → separate conforming OPC UA endpoint	Ignition tags, controls, alarms, trends, and history; mutating controls are enabled only after the gates in Section 6.3

GIZMO-SC-005 SC shall treat the Kria model as the authoritative GIZMo OPC UA contract and shall support GIZMo Kria and legacy GIZMo as independent producers. Platform-specific capability differences shall be represented by status and capability policy; they shall not change canonical NodeIds, datatypes, ranks, units, ranges, or meanings.

GIZMO-SC-006 The ZedBoard server shall neither connect to nor proxy the Kria server and shall not start, stop, configure, or otherwise control it. SC shall use distinct endpoint, application, and device identities for the two connections.

3 Physical and Electrical Interface

3.1 Measurement boundary

The installed measurement path consists of detector ground, building-ground reference, the stimulus path, saturable-inductor assembly, current transformer (CT), analog front end, and the associated cable shields and chassis bonding. The system drawing places the GIZMo, CT, saturable inductor, alarm beacon, and audio indication across the detector/building-ground monitoring boundary.

3.2 External connectors

Connection	Function	Interface rule
Stimulus Out / To Detector	AC stimulus-current driver output	Route only by the approved chassis drawing. Its center and shield are not generic isolated meter terminals.
CT In / From CT	Returned current-transformer signal into the amplifier/ADC	Connect to the CT output with the specified coaxial cable and polarity.
Monitor BNCs	Stimulus-current and CT-output observation	Oscilloscope diagnostics only; loading and grounding shall be controlled.
Detector / building ground	Quantity being monitored	High-current and protective bonding remain governed by the electrical design, not this OPC UA interface.

12-V input	Chassis/control power	Supply, fuse, connector, and power budget shall follow the hardware record.
Ethernet	SC network interface	Copper CAT5e/CAT6 or approved optical path to the restricted controls network.
USB, UART, JTAG	Local commissioning and recovery	Maintenance-only; not a production SC command path.
Beacon/audio	Local physical alarm outputs	Owned by GIZMo/DPS design; SC shall monitor rather than independently drive them unless separately approved.

The legacy rear centers labelled `TODETECTOR` and `FROMCT` are active stimulus/output and signal/input conductors. A powered resistance-mode measurement across them injects meter current into the active loop and is not a passive detector-ground measurement. The observed 50-kohm external-box response is useful characterization evidence but is not a calibration point.

3.3 Kria hardware layer

GIZMo Kria uses an AMD/Xilinx Kria K26 SOM mounted on the KR260 Robotics Starter Kit carrier, the custom control board, FPGA overlay, ADC/DAC signal path, relay drivers, 5-inch EVE display, alarm/audio connections, and two Processing-System Ethernet interfaces. The maintained software maps the installed FPGA resources at fixed addresses; the overlay identity and expected devices are monitored through the public model.

GIZMO-SC-013 The Kria overlay, control-board revision, calibration tables, and physical cable assignment shall form one controlled device configuration.

3.4 Legacy hardware layer

The legacy platform consists of a ZedBoard, 2017 FPGA bitstream, amplifier board 176962, PMod calibrator 176971, and VM801P/FT801 local display. The calibrator has five populated binary-weighted sections measured during characterization as 8.15, 16.17, 31.60, 63.52, and 127.12 ohm. The R6 footprint is unpopulated and participates in the high-impedance disconnect topology. These DMM readings are as-built engineering observations, not traceable standards.

The relay controller accepts a one-shot set/reset pulse for each latching relay. Finite states select combinations of R1–R5; the restored normal state opens the chain through the unpopulated R6 position. Any diagnostic that owns relay MMIO shall first stop `GIZMO.elf`, guarantee cleanup on normal exit, error, and interruption, verify restoration, and then restart the original controller.

4 Link, Transport, and Internal Protocols

4.1 Controls network

The SC-facing physical network is Ethernet. Each OPC UA server shall use a site-assigned address on the restricted controls network. Final hostname, address, route, switch port, VLAN, and firewall entries are installation data.

4.2 Kria private interfaces

Interface	Default	Purpose and policy
ZMon stream	TCP 5055	Private measurement-engine result stream used by the display and command bridge.
ZeroMQ command API	TCP 5555	Compatibility command API for threshold, averaging, calibration, ADC capture, time, and latch actions. It is not the preferred public SC interface.
Temperature stream	TCP 5005	Private temperature producer.
SDR stream	TCP 5556	Private raw acquisition stream; the OPC UA snapshot is for modest-rate inspection, not continuous waveform distribution.
Privileged helper	Unix socket	Root-owned, allow-listed operations including measurement-engine restart and setting the system/RTC time.
OPC UA	TCP 4840	Supported public monitoring and control interface.
Dashboard	HTTP 8080	Read-only operator/engineering view.

The package target `gizmo.target` is the local service-lifecycle boundary. Systemd starts, stops, supervises, and restarts the owned components. Service state is monitored through OPC UA; arbitrary `systemctl` execution is not an OPC UA capability.

4.3 Legacy private interfaces

The 2017 controller is launched by its SysV service in normal measurement mode. Its command parser has native commands including `RUN`, `CAL`, `set_th`, `read_adc`, `set_time`, `set_res`, and `set_rly`. The SysV wrapper supports start, stop, restart, and status. These facts establish native capability; they do not by themselves authorize remote SC control.

The Zed-to-VM801P display link is a private 16-bit SPI exchange at 4 MHz, mode 0, MSB-first. The controller sends a tagged threshold, tagged resistance, and an untagged comparison value; the display returns alarm bits used by the legacy controller. This SPI link is not an SC interface.

The target-local recovery collector reads coherent completed `RUN` frames from the exact executable and retains two bounded journal segments across client or network loss. It checks one PID, executable identity, coherent finite I/Q/magnitude/resistance, and the recovered transfer function before publishing a new sequence. Repeated polls of the same frame do not manufacture new measurements. The conforming server shall consume only validated complete records.

5 Public OPC UA Application Interface

5.1 Normative information model and endpoint identity

The public protocol is OPC UA Binary over UA TCP. Each producer uses a separately assigned endpoint:

```
GIZMo Kria:   opc.tcp://<assigned-kria-host>:4840
Legacy GIZMo: opc.tcp://<assigned-zed-host>:<assigned-port>
```

The namespace URI is `urn:fnal:gizmo`. The generated Kria artifact `schema/gizmo-opcua-contract.json`, model 1.3.1, is the normative machine-readable contract. Its digest is checked by the Kria, ZedBoard, and Ignition repositories. Consumer or ZedBoard capability profiles are non-authoritative and shall contain no contract overrides.

Clients shall resolve the namespace URI at connection time and shall not hard-code a namespace index. Canonical objects are rooted at `Objects/GIZMo`; deterministic string NodeIds include:

```
GIZMo.Identity.*
GIZMo.Measurement.*
GIZMo.Alarm.*
GIZMo.Configuration.*
GIZMo.Operations.*
GIZMo.Calibration.*
GIZMo.Health.*
```

The current released information-model baseline is version 1.3.1, published 14 August 2026. Rows labelled “REQUIRED MODEL 1.4” in the intake workbooks are forward requirements under review; they are not part of model 1.3.1 and shall not be advertised under that version. The workbooks are selected SC integration profiles and do not create independent NodeIds or semantics. If a workbook entry and normative prose conflict, the conflict shall be resolved by joint GIZMo–SC change control before implementation or commissioning.

The model-1.3.1 machine-readable contract contains 43 objects, 457 variables, and five methods. Its embedded canonical contract digest is:

```
contract_sha256=25
   d758ecb65db09a03f5acbbf39db714caa711efbac4d4b286c3ff63b2459201
file_sha256=1a09ec04deb03355a1603e7c2c1279511e520906386981aab064cd9a629afe5a
```

That generated artifact is both the normative machine-readable contract and a reproducible conformance fixture. The maintained Kria implementation is the contract authority; this ICD is its human-facing requirements and change-control companion. Reviewed changes shall be generated from the Kria model and synchronized to consumers and conforming-producer profiles.

Model 1.x revisions preserve NodeId, node class, datatype, rank, units, method signature, and physical meaning. A breaking change requires a new major- version namespace. A shared model revision requires joint GIZMo–SC ICD change control; no conforming producer or consumer may unilaterally publish one. A platform- specific extension, if ever required, shall use a different namespace and shall not be required by SC. The recovered `SimpleOPCUAServer/CommandObject` tree may remain outside the SC boundary for migration, but new SC integration shall use only the canonical model.

GIZMO-SC-016 A GIZMo producer claiming conformance shall match this contract's canonical browse paths, NodeIds, node classes, datatypes, ranks, units, EngineeringUnits/EURange metadata where defined, method signatures, StatusCodes, and source-timestamp meanings for the model version it reports. Its effective access shall be no more permissive than the contract and shall reflect the authenticated role and advertised device capability. A producer shall not report a model version that has not been released through ICD change control.

GIZMO-SC-017 Canonical nodes required by the producer's reported model version shall remain discoverable when a hardware implementation cannot supply their values. Unsupported or unavailable data shall carry `BadNotSupported`, `BadDataUnavailable`, or another specific non-good StatusCode; unsupported methods shall fail with `BadNotSupported`. A platform limitation shall not be hidden by deleting, retyping, renaming, or silently repurposing a canonical node.

GIZMO-SC-018 Address-space conformance does not grant control authority. SC shall enable a write or method only when this contract defines it, the producer advertises the capability, the authenticated role permits it, and the operation passes the separate security, serialization, non-interference, readback, rollback, and commissioning gates in this ICD. The ZedBoard development profile may expose only the bounded `Configuration.ThresholdOhm` write described in Section 6.3; production use remains disabled until the identity, security, non-interference, readback, and commissioning gates pass.

The ZedBoard implementation shall expose the same required baseline nodes and canonical metadata. Unsupported values remain present with the most specific non-good StatusCode; unsupported methods return `BadNotSupported`. A local acceptance limit may reject a write more narrowly than the canonical engineering range, but it shall not rewrite that range in the address space or in Ignition.

5.2 DataValue and quality semantics

Every variable shall provide its OPC UA datatype, StatusCode, and source timestamp. Unknown numeric values shall not be exposed as plausible zero. Before a first sample, the status is `BadWaitingForInitialData`; a retained sample after loss may be `UncertainLastUsableValue`; an unavailable platform feature is `BadNotSupported` or `BadDataUnavailable`. `HIGHZ` is not unavailable data: it is a deliberate good-quality range state.

GIZMO-SC-020 `ResistanceOhm` is a physical value only when `ResistanceRange=InRange`. HIGH Z shall be an explicit range state with a null/NaN resistance and `Good` quality, never a fabricated 500- or 999-ohm measurement. `ResistanceRange=OutOfRange` carries the measured meaning; no source or data-quality fault shall be inferred from HIGH Z alone.

GIZMO-SC-021 `Alarm.Active`, when available, shall come from the device's authoritative alarm branch. SC, the OPC UA producer, and Ignition shall not recompute it from the resistance and threshold readbacks.

5.3 Monitoring groups

The canonical model contains Identity, Measurement, Alarm, Thermal, Time, OperatingSystem, Network, Storage, Firmware, Services, Calibration, SDR, Configuration, Operations, and Health groups. The two platform intake lists are the detailed selected SC integration catalogs: they contain the production-facing variables, controls, command readbacks, and requested future model additions defined by this ICD, rather than all native diagnostic leaves. Every node in the authoritative required baseline remains present on a conforming ZedBoard endpoint and carries a non-good StatusCode when its value is unsupported. Requested future additions do not become canonical until the Kria authority publishes them in a reviewed model revision.

5.4 Numeric-field convention

In the companion intake spreadsheets, *Sensor Span* is a verified software bound or documented presentation domain, *Nominal Value* is the requested initial value or a clearly identified current observation, and *Desired Range* is the initial normal or commissioning check band. A wider authorized write range, where one exists, is stated in Notes. The Low/High, delay, deadband, and action columns define Ignition alarm configuration only. A blank cell means that no supported number or action has been established; it is not zero and it is not "not applicable" data copied from the intake template. Notes distinguish implemented bounds, current device/package values, measured observations, and provisional commissioning choices.

6 Configuration, Control, and Reset Semantics

6.1 Control classification

Operation	Kria native / OPC UA	Zed native	Zed native OPC UA
Set alarm threshold	Yes / read-write	Recovered controller word, persistent file, and display transaction	Canonical read-write metadata; development profile accepts 0–1023 ohm
Set averaging / reads	Yes / read-write	Fixed RUN implementation; launch argument controls cadence, not the same averaging semantics	Fixed readback; write returns `BadNotSupported`
Clear alarm latch	Yes / `ClearLatch`	No equivalent latch protocol in the legacy display/controller path	Method returns `BadNotSupported`
Restart measurement engine	Yes / local allow-listed helper; not presently a canonical method	SysV `restart`	`BadNotSupported`
Reboot controller	Linux capability / not a canonical method	Linux capability	Not exposed
Capture ADC	Yes / `CaptureAdc`	`read_adc`	`BadNotSupported`
Normalize magnitude	Yes / method supported	No validated equivalent	Not supported
Set system time	Yes / `SetSystemTime`	`set_time`; target clock is automatically synchronized	`BadNotSupported`; automatic service live
Calibration sweep	Yes / `StartCalibration`	`CALN`	Maintenance- only; `BadNotSupported`
Direct relay/resistor state	Native maintenance capability	`set_res`, `set_rly`, or reviewed diagnostic	Never a routine SC control

GIZMo Kria supports the controls listed below. The ZedBoard conformance profile permits only a bounded threshold write, subject to commissioning; all other legacy variables remain read-only and all legacy operation methods return `BadNotSupported`. This is a server capability boundary, not a claim that the hardware lacks additional local maintenance functions.

6.2 GIZMo Kria controls

GIZMo Kria exposes two writable UInt32 configuration variables. Their stable NodeIds and ranges are:

```
GIZMo.Configuration.ThresholdOhm              0 .. 1,000,000 ohm
GIZMo.Configuration.AveragesPerCalculation    1 .. 1,000,000
```

The Kria package initializes the threshold and averaging count to 100. For initial joint commissioning, SC shall request 100 ohm and verify the live `Measurement.ThresholdOhm` readback. This value matches the sustained setpoint in the retained Kria history; a brief 2-kohm commissioning excursion is test evidence and is not a desired operating value. Although the firmware accepts zero, the initial Ignition operator range is 1–500 ohm because zero effectively disables the resistance-threshold branch. The accepted OPC UA bounds remain unchanged.

It exposes the following typed methods and command-result variable:

```
GIZMo.Operations.ClearLatch()
GIZMo.Operations.StartCalibration(UInt32)
GIZMo.Operations.CaptureAdc()
GIZMo.Operations.NormalizeMagnitude()
GIZMo.Operations.SetSystemTime(DateTime)
GIZMo.Configuration.LastCommandResult
```

GIZMO-SC-030 SC writes and methods shall be bounded, role-authorized, logged, and followed by readback of the resulting state. A transport-level success alone does not prove that the requested physical state was reached.

GIZMO-SC-031 Clearing the latch shall not suppress an active physical alarm. A device restart or reboot shall not be described as “clear latch.”

GIZMO-SC-032 Restarting `gizmo-zmon.service` and rebooting the Kria are local maintenance/recovery actions until separately specified as canonical OPC UA methods. SC shall not infer those permissions from writable service-status nodes; service-status nodes are read-only.

6.3 Legacy GIZMo conformance and controlled expansion

The ZedBoard OPC UA server is a separate conforming implementation under development. A target-side ARMv7 development build was exercised on 13 August 2026; this does not constitute a production deployment or acceptance. The server design consumes only the target-local recovery journal, requires the controlled executable identity, preserves source timestamps, and remains independent of the Kria endpoint. OPC UA client or controls-network loss shall not stop legacy acquisition. Anonymous sessions are read-only. An authenticated session may receive write access only to `GIZMo.Configuration.ThresholdOhm`; address-space mutation is denied and all operation methods return `BadNotSupported`.

The OPC UA process shall not compete with `GIZMO.elf` for general FPGA, MMIO, relay, or display ownership. It shall not expose arbitrary shell text, file paths, register values, or raw relay states. The sole development exception is the separately reviewed, serialized, and bounded threshold transaction described below.

The recovered executable has one threshold word, one persistent threshold file, and a deterministic display update transaction. The authoritative contract retains UInt32 read-write metadata and a 0–1,000,000-ohm engineering range. The ZedBoard implementation accepts only the hardware-supported 0–1023-ohm subset and shall return `BadOutOfRange` above 1023. Before changing anything its setter requires a fresh journal record, exact controller identity, a running or sleeping process, and agreement among the journal, persistent file, and live word. It briefly stops the controller, atomically persists and reads back the value, updates and reads back controller memory, sends the display transaction, and resumes the same PID. Failure triggers rollback before resume. SC shall never pass arbitrary shell text, file paths, or raw MMIO values to the target.

An authenticated 100-ohm development write passed OPC UA, persistent-file, live-word, recovery-journal, and display-transaction readback without changing the controller or recovery process identity. A wrong password and anonymous write were rejected. The physical display remains an operator-attested acceptance observation; the server cannot see its pixels. This is technical evidence for the candidate design, not release of a production control path.

Stage	**Candidate capability**	**Release gate**
0	Monitoring	Development server and recovery buffering exercised; 100 coherent cycles must still match the local display and complete restart, stale, wrong-hash, partial-frame, and network-loss acceptance.
1	Measurement-engine restart/status	Exact target identity; serialized service ownership; timeout; post-restart PID/hash/freshness readback; audit.

2	Threshold write	Development implementation has authenticated 0–1023 accepted subset, exact identity/coherence gate, persistent/live/display readback, and rollback. The 100-ohm idempotent technical test passed; production server packaging, Ignition role, and physical-display acceptance remain joint commissioning items.
3	ADC capture and clock correction	Output-file ownership, bounded transfer, timeout, target-time quality, and restoration of normal `RUN` mode tested.
4	Calibration sweep	Approved isolated fixture; exclusive relay ownership; raw I/Q and state provenance; guaranteed HIGH Z cleanup; independent validation; no automatic activation.

GIZMO-SC-033 A failed, timed-out, or interrupted legacy command shall leave the last verified normal measurement mode or report a latched command-failure state requiring maintenance. It shall not report success from command submission alone.

GIZMO-SC-034 Direct `set_res` and `set_rly` shall remain expert, isolated maintenance operations. They shall not be general-purpose Ignition buttons or writable tag values.

GIZMO-SC-035 A future restart method shall name its scope explicitly, for example `RestartMeasurementEngine`; a generic `Reset` method is prohibited.

7 Detector Control Display, Alarms, and History

7.1 Display

Ignition shall show source identity, endpoint health, sample age, measurement quality, resistance/range, threshold, calibration state, and relevant control capabilities. The hardware implementation shall be labelled *GIZMo Kria (K26/KR260)* or *Legacy GIZMo (ZedBoard)* wherever it could be ambiguous. HIGH Z is rendered as a state, not as a precise endpoint. The local display remains an independent commissioning reference; disagreement creates a diagnostic event and never triggers automatic calibration.

7.2 Sampling and freshness

Kria measurement data update nominally each second; platform data update at 10- or 30-second cadence. Clients should use subscriptions. The legacy recovery collector may sample each second, but it emits a record only for a new coherent completed `RUN` result. The conforming ZedBoard server shall publish that source sequence, and Ignition shall not manufacture intervening history samples. A stale source becomes non-good within two expected completed cycles.

7.3 Ignition history boundary

Ignition shall subscribe directly to each GIZMo OPC UA endpoint and shall own the production tag history. History shall preserve the reported device identity, OPC UA source timestamp, StatusCode, explicit HIGH Z states, and command audit events. The SC integration requires no intermediate history service between OPC UA and Ignition. Ignition history failure shall not stop local measurement or physical alarm indication.

7.4 Alarms and Detector Protection

SC shall alarm separately on physical ground-impedance alarm, communication loss, stale data, non-good measurement quality, identity mismatch, and command failure. An unavailable legacy `Alarm.Active` remains unknown; it shall not be converted to clear/false.

The following values are the initial SC configuration. A provisional operating envelope is a rounded commissioning check band derived from retained or live observations; it is not automatically an alarm threshold. Such values may be revised through change control without changing the OPC UA contract.

Parameter	GIZMo Kria	Legacy GIZMo	Basis/status
Resistance presentation domain	0–500 ohm; higher values are HIGH Z	0–500 ohm documented display domain; higher values are HIGH Z	Kria implemented; legacy metrology still requires installed-loop validation
Threshold range and policy	Canonical UInt32 engineering range: 0–1,000,000 ohm; initial operator range: 1–500 ohm	Same canonical metadata; ZedBoard accepted subset: integer 0–1023 ohm; initial operator range: 1–500 ohm	Authoritative contract plus platform capability and initial policy

Initial threshold	100 ohm; package default and sustained retained-history value	100 ohm; confirmed in direct frames, adapter-pilot cycles, and native authenticated write/readback	Requested initial setpoint
Resistance comparison	Alarm when resistance is strictly less than the live threshold	Alarm-return bits assert when rounded displayed resistance is strictly less than threshold	Recovered device behavior
Additional Kria phase alarm	If resistance is greater than 8 ohm and the absolute interpolated-versus-`atan2` phase difference exceeds 1.5 degrees	No validated equivalent	Implemented Kria logic
Ignition physical-alarm trigger	`Alarm.Active=True`, 0-s on-delay, no additional SC hysteresis	Disabled until actual return-word readback is exposed and validated; then the same Boolean trigger/timing applies	Initial requested SC configuration
Automatic action	Annunciate and direct operator response; no automatic power or protection command	Same	Initial interface scope

The Kria measurement engine contains no alarm on-delay or hysteresis in the reviewed decision path. The recovered legacy display comparison likewise has no delay or hysteresis. SC therefore adds neither: the Ignition on-delay is 0 s and the deadband field is blank. Ignition shall alarm on the Boolean `Alarm.Active`; it shall not create a second resistance comparator that could disagree with the phase-aware Kria decision or the legacy rounded-value decision.

The retained Kria SQLite replica covers 27 July through 3 August 2026 and contains 560,762 fast samples, 56,082 platform samples, 9,457 minute rollups, and 227 transition events. It confirms a sustained 100-ohm threshold, a stable 1436-Hz stimulus value, HIGH Z as the predominant normal range state, and both implemented alarm-reason branches. Its steady-state distributions are used to populate provisional Desired Range fields for temperatures, utilization, available memory, phase, and lock-in counts. Commissioning and startup excursions are retained in Notes but are not promoted to desired values. The retained run was captured under model 1.3.0 and the then-installed runtime 0.4.4; the maintained public source baseline reviewed for this ICD is runtime 0.4.6 and model 1.3.1.

A read-only live check of the connected legacy board on 12 August 2026 collected 15 coherent completed `RUN` frames from the unchanged `GIZMO.elf`. The controller returned a 100-ohm threshold, 15 accumulations, 111.554–111.902 ohm calculated resistance, and a 112-ohm green display. The I, Q, magnitude, and derived `atan2` observations are recorded in the legacy workbook as same-setup commissioning envelopes. They do not validate the missing CT/inductor path or the full detector-loop calibration.

The deployed adapter subsequently captured 27 contiguous cycles (`Sequence` 21–47) on 13 August 2026. Resistance was 111.465703–112.010978 ohm (mean 111.722419 ohm), the threshold remained exactly 100 ohm, and the calculated legacy display-equivalent rounding was 112 ohm in 26 cycles and 111 ohm in one. Target communication was then lost before 100 cycles. The captured values carried `UncertainLastUsableValue` and `Measurement.Quality=Uncertain`, consistent with the unvalidated legacy measurement path; after the loss, the adapter exposed `BadNoCommunication` and `Health.Overall=Failed` rather than fresh data. This adds evidence for the attributes and loss semantics but does not close **OPEN-05**.

The target-side ZedBoard development server was then exercised on 13 August 2026. Its authenticated idempotent write of the unchanged 100-ohm setpoint passed OPC UA, persistent-file, live controller-word, recovery-journal, and display- transaction readback. Anonymous and wrong-password writes were rejected, the averaging readback remained the fixed value 15, and every operation method remained `BadNotSupported`. This supplies technical threshold-path evidence, but does not release the server or close the 100-cycle physical-display acceptance in **OPEN-05**.

No reviewed GIZMo or SC source establishes numerical temperature, CPU, memory, storage, or similar supervisory alarm thresholds. Observed envelopes therefore remain monitor/trend checks and the spreadsheet alarm cells remain intentionally blank. Approved limits can be added without changing the OPC UA NodeIds.

Priority, response ownership, and any Detector Protection action still require joint approval by the GIZMo owner, SC, DPS, and operations. A required hard protective action shall not depend only on Linux, OPC UA, or Ignition.

8 Calibration and Maintenance

8.1 GIZMo Kria

GIZMo Kria calibration is a controlled maintenance method, not an automatic SC reaction. Production tables shall be staged, validated, content-hashed, activated atomically, and rollback-tested. Success, error, timeout, process termination, and power-interruption paths shall restore and verify the normal relay state. Calibration state and the active table identities are published through OPC UA.

8.2 Legacy GIZMo

The legacy `CALN` command selects 32 R1–R5 states and logs lock-in measurements. It does not fit, validate, persist, or activate a transfer function; it does not use the abandoned `calib2.txt`; and it does not guarantee normal-state restoration at completion. Starting `RUN` restores the normal `SetRes(-1)` selection, but an interrupted calibration can leave a finite test resistance selected.

Accordingly, a conforming ZedBoard endpoint shall report `Calibration.State=CharacterizationOnly` or `Unvalidated`. A future calibration workflow requires the restored CT/saturable-inductor fixture, measured standards, a common runtime/calibration acquisition primitive, raw I/Q and relay-state records, up/down sweeps, blind validation, clipping and hysteresis checks, guaranteed cleanup, atomic model activation, and rollback. Capacitance remains unsupported until mixed R/C characterization establishes a defensible model.

9 Security and Access Control

Every GIZMo endpoint belongs on the restricted controls network or behind approved authenticated transport. Final certificates, trust lists, usernames/roles, and firewall rules are installation data jointly assigned by SC and site cybersecurity.

GIZMo Kria currently supports an explicitly configured insecure OPC UA policy for an isolated network; serialization is not encryption. Production deployment shall document whether that exception is accepted or deploy the site certificate and role policy.

The ZedBoard development build is restricted to an isolated maintenance interface. A username/password token used with SecurityPolicy None is plaintext on the wire and shall not be routed or bridged to an untrusted network. Any production credential shall be supplied to Ignition through its protected connection credential store and requires joint security review. Unrestricted root SSH is maintenance access, not a production command interface. The conforming server profile allows only the typed bounded threshold write with identity, coherence, readback, audit-result, and rollback gates. It does not expose arbitrary command strings, environment values, file paths, shell execution, general FPGA MMIO, or relay bit patterns.

Credentials, private keys, and unbounded log text shall not be published as OPC UA values or Ignition history/audit events.

10 Responsibilities

Area	GIZMo system owner	SC / other group

Physical system	Supply, label, install, maintain, and document the chassis, analog chain, CT/inductor, calibrator, display, power, alarms, and cables.	I&I provides approved routing and installation support; SC does not alter the analog or ground wiring.
Device software	Maintain the authoritative Kria runtime/model, Kria overlay, preserved Zed image, and conforming ZedBoard server; define meanings, control effects, safe states, and rollback.	SC supplies the integration framework and endpoint host resources where applicable.
OPC UA	Publish the contract from the Kria model; implement and test each producer/server and its readbacks without platform-specific contract overrides.	SC integrates two independent connections, tags, quality, certificates, roles, alarming, and displays in Ignition.
Controls	Define typed operations, bounds, state transitions, and cleanup.	SC implements role-authorized operator workflow and audit; operations approves production use.
Calibration	Own standards, method, fits, validation, activation, rollback, and relay cleanup.	SC displays state/history and provides authorized workflow; it does not infer validity.
History	Publish device identity, quality, source time, model version, and command audit through OPC UA.	SC configures Ignition tag history, retention, restore, and operational access.
Protection	Define hazards and required physical response.	DPS owns reviewed hard protection; SC owns supervisory alarm presentation.

11 Test Stand, Commissioning, and Acceptance

11.1 Physical and common interface acceptance

Before SC sign-off, the parties shall:

- verify connector labels, power, fusing, shield/ground assignment, CT polarity, and the approved stimulus path;
- resolve the namespace URI and verify every required NodeId, datatype, access class, EngineeringUnits value, StatusCode, and source timestamp with a generic OPC UA client;
- verify DeviceId, platform, runtime/ELF identity, endpoint, and time;
- force disconnect, stale data, restart, unsupported-value, and out-of-range-write conditions and confirm non-good quality and last-good semantics;
- confirm HIGH Z is persisted as `ResistanceRange=OutOfRange` with `Good` quality and never as a fabricated numeric value;
- verify Ignition display, alarm journal, command audit, trends, and history identity separation; and
- demonstrate that OPC UA or Ignition-history failure does not stop local measurement or physical alarm indication.

The same versioned, machine-readable conformance checks shall be run against every GIZMo producer. They shall compare the namespace URI, browse paths, NodeIds, node classes, datatypes, ranks, units/ranges, method signatures, access/capability restrictions, quality behavior, source timestamps, and instance identity against this ICD. Passing those tests establishes interface conformance only; it does not establish metrology or authorize controls.

11.2 GIZMo Kria acceptance

All maintained runtime tests shall pass. Threshold and averaging writes shall enforce type and range; every method shall report a bounded result. Latch clear, time correction, ADC capture, magnitude normalization, and calibration shall be tested with authorization and readback. Calibration testing shall include error/interruption cleanup and rollback. A 24-hour soak and power-interruption recovery test shall complete before production approval.

11.3 Legacy GIZMo monitoring acceptance

At least 100 completed cycles shall match the physical display's rounded resistance, including identical consecutive readings. Wrong hash, multiple or changed PID, target restart, server restart, network outage, partial update, stale frame, and non-finite values shall not produce a false fresh sample. Anonymous writes, every non-threshold variable write, and every method shall fail explicitly. Ignition shall retain the ZedBoard samples with their reported device identity, source timestamp, and StatusCode.

11.4 Legacy GIZMo control acceptance

Each control stage in Section 6.3 requires a separate test record. Threshold acceptance shall cover anonymous, wrong-credential, wrong-type, out-of-range, stale/mismatched-state, allowed, repeated, and rollback cases plus persistent, live-word, journal, OPC UA, and physical-display readback. Future stages shall additionally cover simultaneous requests, controller crash, timeout, power interruption, audit completeness, safe-state restoration, and rollback to the capability-bounded release. Calibration and relay tests additionally require an approved isolated electrical setup and independent observation of the restored HIGH Z state.

12 Open Items and Change Control

ID	Open item / release gate	Owner
OPEN-01	Assign EDMS identifier, reviewers, approvers, and controlled distribution.	GIZMo + SC
OPEN-02	Assign production endpoint hostnames/IPs, switch ports, certificate policy, trust lists, roles, and firewall rules.	SC + cybersecurity
OPEN-03	Review the requested 100-ohm threshold and zero added delay/no-hysteresis configuration; approve final priority, response owner, and any DPS action.	GIZMo + SC + DPS
OPEN-04	Add explicit Platform, AcquisitionAdapter, ValidatedResistanceMaximumOhm, and capability nodes in a backward-compatible model revision.	GIZMo software
OPEN-05	Complete the conforming ZedBoard server implementation, publication-safe packaging, and 100-cycle physical-display acceptance.	GIZMo + SC
OPEN-06	Expose and validate OPC UA readback of the actual legacy alarm-return word used by the controller. The bit meanings and physical SPI exchange have been recovered; the server shall not synthesize this readback from resistance.	GIZMo firmware
OPEN-07	Define, implement, and test any additional allow-listed legacy controls beginning with an explicitly scoped measurement-engine restart; threshold control shall remain the only candidate write until then.	GIZMo software + SC
OPEN-08	Decide whether a canonical `RestartMeasurementEngine` method is required for both platforms; do not add a generic reset command.	GIZMo + SC + operations
OPEN-09	Restore the CT/saturable-inductor fixture and validate legacy resistance/phase calibration with measured external standards.	GIZMo hardware

Any change to a connector, electrical reference, NodeId, datatype, physical meaning, access class, command effect, alarm ownership, calibration authority, safe state, or historized-field set requires joint review and a new controlled revision, including review of source identity, quality semantics, and retention.